Lesson 4 Teacher's Key

Scratch

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

Two scripts on two sprites - the Bullet's existing "when I start as a clone" forever loop gets one new if-branch, and the Enemy gets one new listener script:

```
-- On the Bullet sprite (added inside the existing forever loop, before
the off-screen check):
when I start as a clone
  show
  go to x: (Player's x) y: (Player's y)
  forever
    if <touching [Enemy]?> then
      broadcast [enemy hit]
      delete this clone
    change y by 10
    if <y position > 170> then
      delete this clone

-- On the Enemy sprite:
when green flag clicked
  go to x: 0 y: 150
  show

when I receive [enemy hit]
  hide
```

Level 2 - Complete Worked Solution

This already works with no extra code: Scratch's "touching" check always returns false for a hidden sprite, so once the Enemy sprite is hidden, no later bullet clone can trigger a second "enemy hit" broadcast, even with several bullets in flight at once.

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: play on hit

Upload sfx_hit.wav from this lesson's Assets folder, then add "start sound [sfx_hit]" to the Enemy's "when I receive [enemy hit]" script, right before hide:

```
when I receive [enemy hit]
  start sound [sfx_hit]
  hide
```

Visual: give the enemy its own look

A Costume editor recolor, or a "set color effect to" block in the Enemy's green-flag script, makes it read clearly as a threat:

```
when green flag clicked
  go to x: 0 y: 150
  set color effect to 50
  show
```

Structure: keep hit-response on the Enemy sprite

The Bullet's forever loop should only ever detect a hit and report it (touching check + broadcast + delete this clone) - everything about how the Enemy responds to being hit (sound, hiding) belongs on the Enemy's own "when I receive [enemy hit]" script. Keeping this boundary clean is what let the Sound idea above drop into the Enemy sprite alone.

Complete Level 3 solution (all 3 goals combined)

Everything above combined, across the Bullet and Enemy sprites:

```
-- Bullet sprite (inside the existing forever loop):
when I start as a clone
  show
  go to x: (Player's x) y: (Player's y)
  forever
    if <touching [Enemy]?> then
      broadcast [enemy hit]
      delete this clone
    change y by 10
    if <y position > 170> then
      delete this clone

-- Enemy sprite:
when green flag clicked
  go to x: 0 y: 150
  set color effect to 50
  show

when I receive [enemy hit]
  start sound [sfx_hit]
  hide
```